

A Machine Learning–Based Factory Reallocation and Shipping Optimization Recommendation System for Nassau Candy Distributor

Applied Research Paper / Technical Research Report

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Abstract

This study presents an applied machine learning framework for identifying slow-shipment risk and screening factory-reallocation scenarios for Nassau Candy Distributor. The analysis uses 10,194 transactional records containing order, product, geographic, shipping-mode, and financial attributes. The raw dataset contains 15 products, 3 divisions, 4 sales regions, 5,044 customers, and 8,549 unique orders.

Delivery_Days has a mean of 1320.84 days, a median of 1274 days, and a third quartile of 1638 days. The binary target Slow_Shipment uses a train-only threshold of 1638 days. An order-level group split produced 8,132 training records and 2,062 test records with zero order overlap.

Three classifiers were evaluated. Random Forest was selected for the operational layer because it achieved the highest F1 score (0.6773) among the candidates while retaining recall of 0.8925 and ROC-AUC of 0.9013. A counterfactual engine generated 75 product–factory scenarios. Because verified factory coordinates and transportation-cost inputs were unavailable, the scenario layer is explicitly treated as predictive screening rather than causal or cost optimization.

1. Introduction

Distribution operations require decisions that connect commercial performance with logistics execution. This project combines delivery-risk prediction, model evaluation, explainability, and factory-reallocation scenario screening into one reproducible workflow. The goal is to identify slow-shipment risk and provide evidence for further operational review without overstating causal effects.

2. Dataset and Problem Definition

The supplied CSV contains 10,194 rows and 18 columns. Total sales are \$141,783.63, gross profit is \$93,442.80, cost is \$48,340.83, and total units are 38,654. The overall gross profit margin is 65.91%. Order dates range from 2024-01-02 to 2025-12-31. Shipment dates range from 2026-06-30 to 2030-06-28. Delivery_Days is calculated from the supplied date fields.

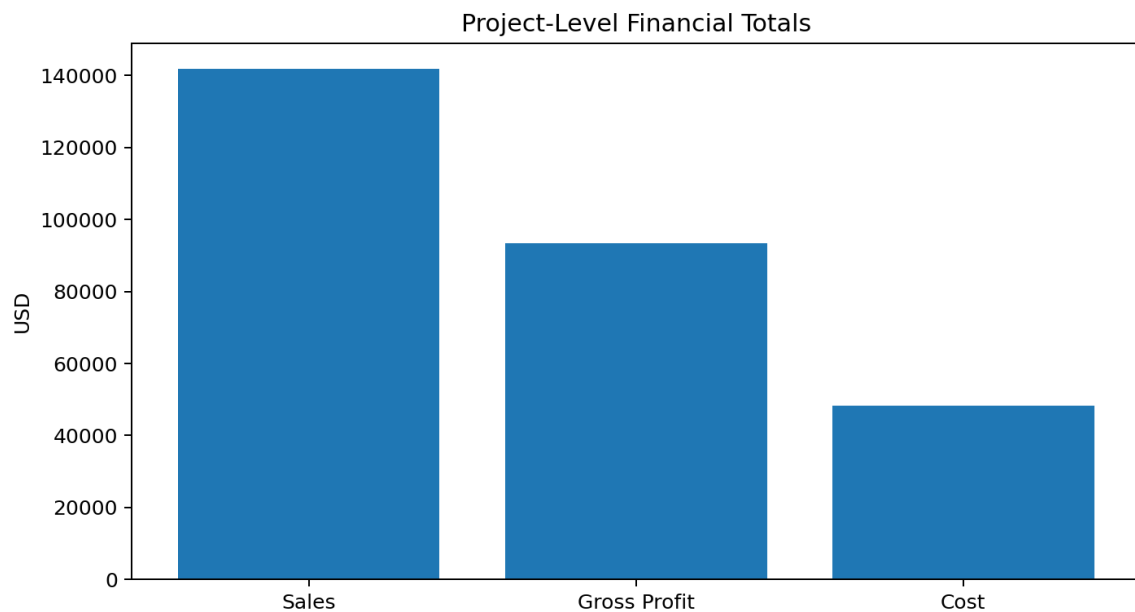


Figure 1. Project-level financial totals.

3. Methodology

3.1 Data validation

Dates were parsed using the observed day-month-year format. Delivery_Days was calculated as Ship Date minus Order Date. The completed audit reported no missing delivery-day values and no negative delivery intervals.

3.2 Target construction

Slow_Shipment = 1 when Delivery_Days exceeds the training-set third quartile of 1638 days; otherwise 0.

3.3 Leakage control

Order ID was used as the grouping variable. GroupShuffleSplit produced 8,132 training records from 6,839 orders and 2,062 test records from 1,710 orders, with zero order overlap.

3.4 Features

Fifteen predictors were used: Product Name, Division, Country/Region, State/Province, Region, Factory, Ship Mode, Units, Sales, Cost, Gross Profit, Order_Year, Order_Month, Order_Quarter, and Order_DayOfWeek. Ship Date, Delivery_Days, and Slow_Shipment were excluded from the model inputs.

3.5 Models

Logistic Regression, Random Forest, and Gradient Boosting were trained in preprocessing pipelines. Performance was evaluated with accuracy, precision, recall, F1, and ROC-AUC.

4. Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.7978	0.5070	0.9322	0.6568	0.8863
Random Forest	0.8235	0.5457	0.8925	0.6773	0.9013
Gradient Boosting	0.8346	0.5868	0.6869	0.6329	0.9043

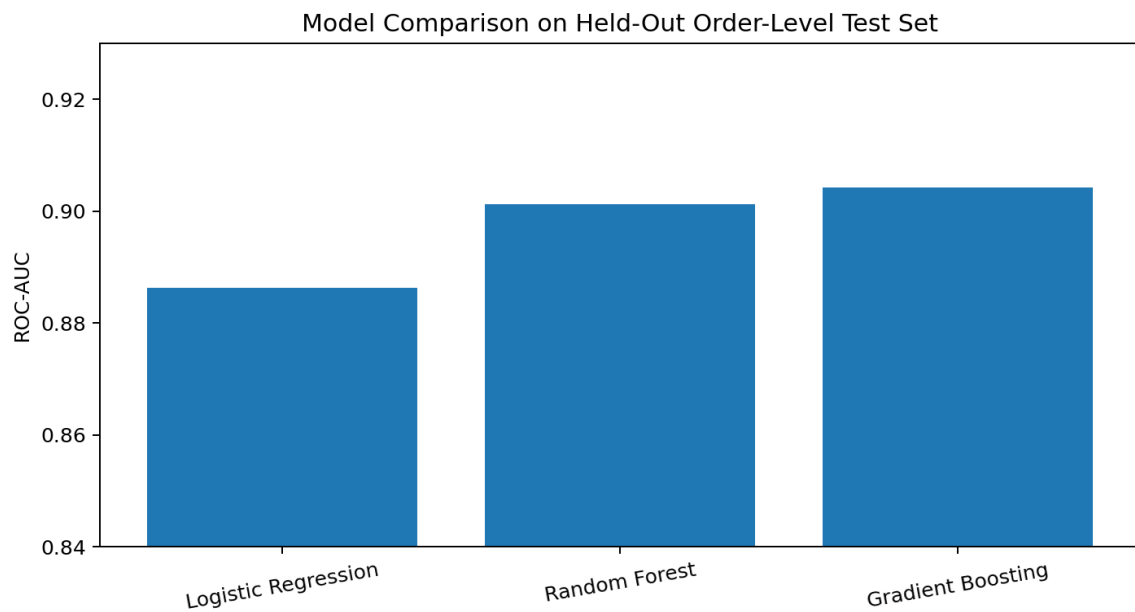


Figure 2. ROC-AUC comparison on the held-out order-level test set.

Gradient Boosting has the highest ROC-AUC (0.9043), while Random Forest has the strongest F1 score (0.6773) and high recall (0.8925). The project therefore selects Random Forest using F1, recall, and ROC-AUC as the decision hierarchy.

5. Factory Mapping and Baseline

All 15 products were assigned to one validated current factory. The mapping used in the project was: Lot's O' Nuts for three Wonka Bar products; Wicked Choccy's for two Wonka Bar products; Sugar Shack for five Sugar/Other products; Secret Factory for three products; and The Other Factory for two products. The mapping is consistent with the publicly available Nassau Candy profitability analysis used as contextual validation. [cite turn0search0](#)

Factory	Products	Records	Orders	Sales	Gross Profit	Margin
Lot's O' Nuts	3	5692	4760	\$76,340	\$52,771	69.13%
Wicked Choccy's	2	4152	3445	\$55,353	\$36,054	65.13%
Secret Factory	3	217	213	\$8,588	\$4,345	50.59%
The Other Factory	2	100	98	\$1,282	\$152	11.87%
Sugar Shack	5	33	33	\$221	\$121	54.86%

6. Explainability

The selected Random Forest was inspected using impurity-based feature importance and held-out test-set permutation importance. Permutation importance measures the degradation in a chosen score after shuffling a feature and is useful as a model-agnostic inspection technique. Held-out evaluation is preferable when the objective is to understand generalization. [cite turn1search0 turn1search1](#)

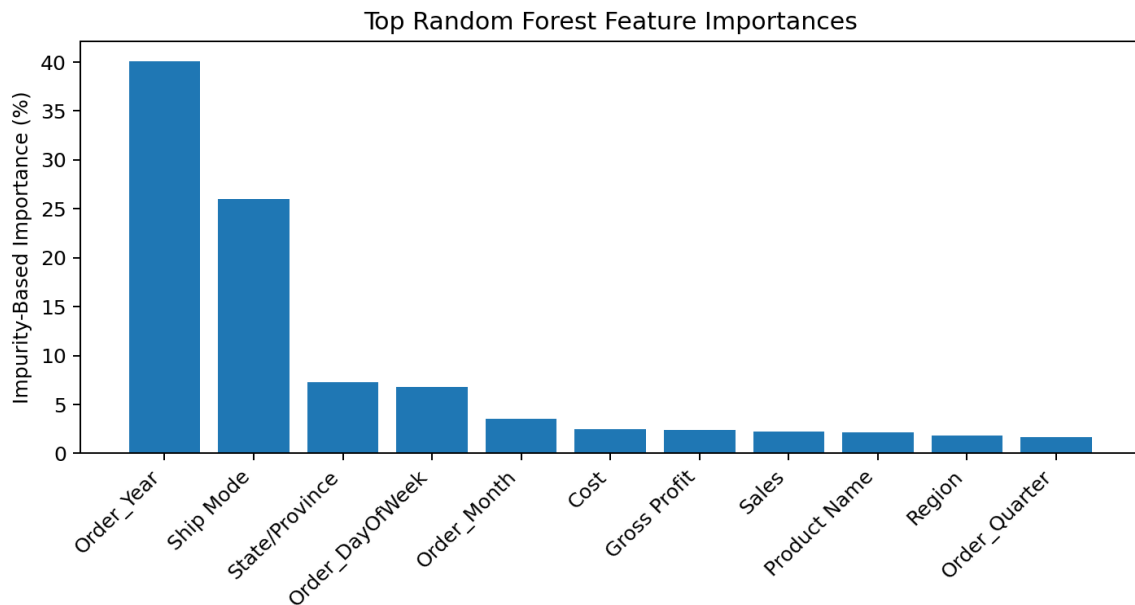


Figure 3. Top Random Forest impurity-based feature importances.

The leading impurity-based signals are Order_Year and Ship Mode. These values describe the fitted model's predictive structure and should not be interpreted as causal effects.

7. Factory Reallocation Scenario Engine

The engine changes only the Factory input and scores every product under all five factory labels, producing 75 scenarios. Probability reduction is defined as Current Slow Probability minus Scenario Slow Probability. A five-percentage-point threshold is used as a conservative screening rule for a strong ML signal. The resulting recommendations are analytical candidates, not guaranteed operational improvements.

Product	Current	Scenario	Current P	Scenario P	Reduction pp
Everlasting Gobstopper	Secret Factory	The Other Factory	0.5247	0.4989	2.59
Fizzy Lifting Drinks	Sugar Shack	The Other Factory	0.2926	0.2782	1.44
Laffy Taffy	Sugar Shack	The Other Factory	0.3659	0.3529	1.30
Lickable Wallpaper	Secret Factory	The Other Factory	0.3451	0.3329	1.22

No transportation-distance or monetary savings are claimed because verified factory coordinates, freight rates, capacity constraints, and carrier data were not available.

8. Discussion

The project demonstrates a complete applied ML workflow in which data leakage is controlled at the order level, the target threshold is defined without test-set information, multiple models are compared, and the selected model is connected to a decision-support scenario engine. Random Forest provides the best F1/recall balance for the stated business objective, while Gradient Boosting provides the best ROC-AUC.

The scenario engine must be interpreted conservatively. A model score for an unobserved product–factory combination is a counterfactual prediction under the model, not evidence of a causal treatment effect. Operational validation through controlled pilots is required before changing factory assignments.

9. Limitations

- The dataset is historical and observational; predictive associations do not establish causality.
- The target is a project-defined slow-shipment classification based on the training-set third quartile rather than a contractual service-level agreement.

- Delivery_Days values are very large in absolute terms because the supplied Ship Date values occur years after the Order Date; the model faithfully analyzes the supplied records rather than replacing them with assumed dates.
- Factory coordinates were not verified, so distance-based routing and transportation-cost optimization were intentionally not performed.
- The counterfactual factory scenarios may represent unsupported combinations in the historical data. They are screening scenarios, not experimentally observed outcomes.
- Impurity-based feature importance can be biased; held-out permutation importance was therefore included as a complementary diagnostic.
- No external transportation cost, carrier capacity, inventory constraints, production capacity, or service-level agreement data were available in the supplied project dataset.
- The model was evaluated on one held-out order-level split. A future production study should add repeated group-aware validation and temporal validation.

10. Conclusion

This applied study developed an audited machine learning system using 10,194 transactions. The selected Random Forest achieved 0.8235 accuracy, 0.5457 precision, 0.8925 recall, 0.6773 F1, and 0.9013 ROC-AUC. The system generated 75 factory scenarios across 15 products and five factory labels, while deliberately avoiding unsupported transportation-cost claims. The work is best viewed as a reproducible research prototype and internship-level decision-support system, with clear pathways for future enhancement through verified logistics and geographic data.

References

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- Nassau Candy Distributor dataset supplied for the applied ML project: Nassau Candy Distributor.csv.